

Unit

Computer Networks

Title

LAN Design for a Corporate Office

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Preface

TechSavvy Inc company which is specialized in developing up-to-date software solutions for various companies such as healthcare, commercial and finance. It develops rapidly and needs to expand its employees. It has a planned office layout and specific network requirements when it is moving to the new office.

Task 1

a) Key requirements for LAN design

The company requires LAN design for 150 employees. Eight departments will be added in this company such as development, finance, human resources, sale and marketing, customer service, research , IT support and administration.

In development, finance, human resources, sale and marketing, customer services and research department, about 20 employees will work in each of these. About 15 employees will work in both IT support and administration.

Switches, desktop computers, VoIP phones, printers, access points and mobile devices are added in each of all departments. To access internet and data storage , modem, routers and servers will be implemented. Video projectors will be added in video conferencing rooms for meeting. To be secure network, firewall must be added into network design.

IT associated departments such as development, research and IT support need high speed internet access for coding, larger file transfers and remote work. Customer service department and sale and marketing also need high speed and sufficient bandwidth for meeting and video conferencing.

Finance, administration and human resources departments need more secure LAN design to protect sensitive data and prevent unauthorized access. Segmentation of network can be used for both security and network traffic.

b) Company's growth plans and requirements for scalability

The company need to expand 300 employees in the future and add new technologies for future growth. Network IP address also should be adjustable for scalability. To avoid network traffic, network needs to be segmented for server rooms, open workspaces, private offices, conference rooms, IT support and break rooms. It is good not only for speed but also for security and privacy enhancement.

For redundancy and reliability, backups should be ready especially for servers. The wireless network also needs to be strong and reliable. In remote work, VPN should be used for secure traffic.

The company uses video conferencing and larger file transfers. They require high-bandwidth applications and services for real time processes. They depend on high-speed internet to maintain their quality.

So, network design needs to be adjusted. The router should closer distance to conference rooms and be good quality for performance. There will be no interference and best for keep in open space. Network traffic also needs to be carefully managed by segmentation and network plan should be with more bandwidth. So, high bandwidth switch and router should be used. (NordVPN, 2025)

VoIP phones are used in company as they are great alternative of line phones. They are low cost and better quality but choppy audio, drop calls and echo can occur when internet is not stable and insufficient bandwidth. So, wired connection (cable) should be used. Security beaches can also occur as it has no encryption and thus firewall must be added in network design.

Task 2

a) Detailed floor plan

Floor (3 floors)	Network devices	Rooms
3 rd floor	- Switch - 6 access points - mobile devices - Desktop computers - VoIP phones - printers	- open plan workspace
		- dedicate spaces (IT support and equipment storage)
2 nd floor	- modem - Firewall - Router - 2 Switch - 4 access points - mobile devices - video projectors - Desktop computers - VoIP phones - printers - Servers	- 15 private offices
		- server room
		- 4 conference rooms
1 st floor	- Switch - 6 access points - mobile devices - Desktop computers - VoIP phones - printers	- open plan workspace
		- common areas (break rooms, cafeteria)

New office layout of the company has 3 floors and layout will be planned in detail as follows.

2nd floor

Rooms

2nd floor will be important floor of company as a server room will be located. Four conference rooms for video conferencing and fifteen private offices for management will have been added in this floor because they are also important. Video conferencing rooms need high speed internet access for their video quality more than other rooms. The private offices and server room also need more security than other rooms. 2nd floor will have 20 employees.

Network devices

Modem will be placed in 2nd floor to modulate and demodulate between internet and router. It acted as a gateway to internet. Firewall will be also placed for network security.

Router will be placed near conference rooms because short, high-quality Ethernet cable can make the maximum speed.

Multilayer switch will be added in this floor for connection of other switch in each floor. Then, another switch will be added which use to connect multiple devices such as computers, printers, access point and VoIP phones.

Access point for wireless devices will be added in each conference room. It will be placed at the central of room and near ceiling to get the strong and reliable wireless network.

Desktop computers, VoIP phones, and printers will be added in conference rooms and private offices for working and effective communication. Video projectors will also be added in conference rooms. In server room, servers only will be placed for security.

1st floor

Rooms

In 1st floor, there will be open plan workspace for software developers and engineers and common areas for breakrooms and cafeteria. They are suitable in this floor and also to each other. 1st floor will have 60 employees.

Network devices

One switch will be added in this floor which use to connect multiple devices such as computers, printers, access point and VoIP phones.

Six access points for wireless mobile devices will be added in this floor. Four access points for open plan workspace and two access points for common areas will be determined. Each access point will be placed central and near the ceiling to get the strong and reliable wireless network.

Desktop computers, VoIP phones and printers will be added in open plan workspace for working and effective communication.

Although there are no wired devices in common areas, ethernet drops will have been installed for future expansion. This will support scalability and minimize the future installation disruptions.

3rd floor

Rooms

In 3rd floor, there will be also open workspace for software developers and engineers and dedicate spaces for IT support and equipment storage. They are suitable in this floor and also to each other. 3rd floor will have 70 employees.

Network devices

One switch will be added in this floor which use to connect multiple devices such as computers, printers, access point and VoIP phones.

Six access points for wireless mobile devices will be added in this floor. Four access points for open plan workspace and two access points for dedicated space (IT support office) will be determined. Each access point will be placed central and near the ceiling to get strong and reliable wireless network.

Desktop computers, VoIP phones and printers will be added in open plan workspace and IT support office for working and effective communication.

Recommendation

By the above facts, physical star topology is suitable for this LAN design. It is also reliable as if one cable fail, others will still work. It is no data collision and so it is high performance. It is less expensive, easy to detect fault and also easy to line up. (geeksforgeeks, 2025)

b) Logical network topology

Overall view

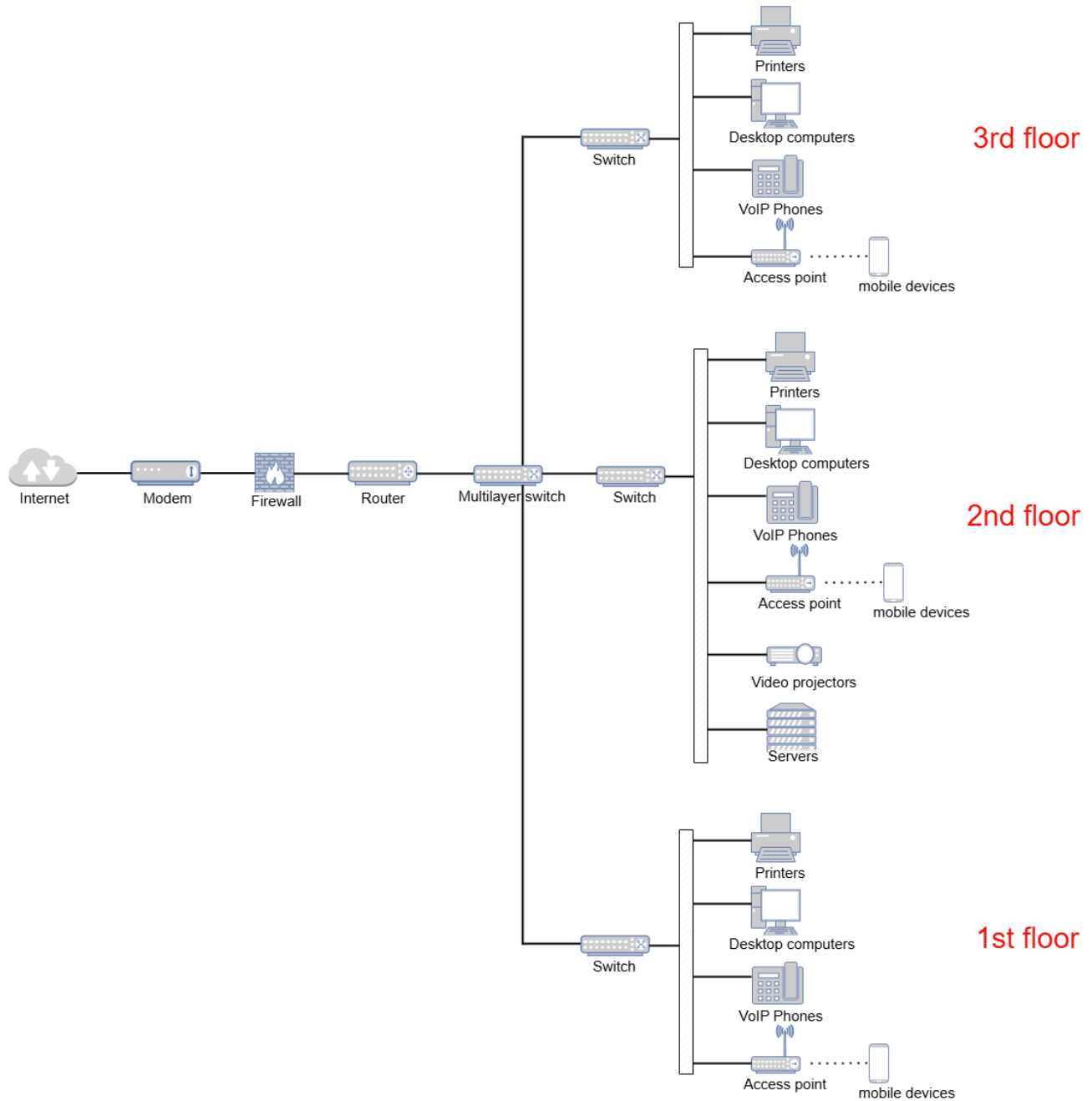


Figure 4. Logical network topology that shows the connections between all network devices

Task 3

a) Necessary network equipment and detailed specifications

Modem

Modem is used to modulate and demodulate of signals. It translates digital 1 and 0 from computer to analog information and has one Ethernet port to one IP address. It works in data link layer (Layer 2). It is a gateway to internet. Digital cable modem (ARRIS SURFboard SB8200) will be used.

Router

Router transmits data from source network system to destination network system. It connects modem to access WAN and has LAN port. It determines route of each device by own IP address. It works in network layer (layer 3) and maintains IP address in routing table. It can be used in wire and wireless network for network segmentation. Edge router (Cisco ISR 4000 Series) will be used.

Switch

Switch connects group of network devices to make LAN. It acts as extension of router as router has few ports. It transmits data between network devices and determines route of each device by own IP address in multilayer switch. It works in data link layer (layer 2) and maintains MAC address in lookup table. Multilayer switch works in both layer 2 and 3. It is used in wired network for network segmentation. Managed switch (Cisco Catalyst 9300 Series) and multilayer switch (Cisco MDS 9100 Series) will be used.

Access point

It is a network device that use to extend wireless coverage of network and distribute signals through wireless. It is a basic system to create wireless LAN and has limited range. It works in data link layer (layer 2) and is used for indoor or outdoor wireless network. Standalone AP (Ubiquiti UniFi U6) will be used.

Ethernet Cable with connector

It is a physical encased wiring over data transfer. It contains 8 twisted cables in 4 pairs and joins with standard connectors which is made of plastic. It is used for connecting network devices and works in physical layer (layer 1). It is used as wiring for Gigabit Ethernet in data center and commercial building. Ethernet cable (Cat6a , 10GBASE-T Ethernet) and RJ 45 connector (T-568B) will be used.

(Gordon, 2024) (Commscope, 2025) (Westermo, 2025)

b) Equipment choices

Modem (ARRIS SURFboard SB8200)

It uses Broadcom BCM3390 chipset and has 3 GB ram, 10 Gbps downstream and 2 Gbps upstream. So, it is high speed and perfect for owner with best internet plan supporting to 2Gbps but need second IP address to get max speed.

It can use IPv6. It is stable performance, future proof and easy to use. It is compatible with major US internet service provider such as Xfinity, Cox.

It has dual Gigabit Ethernet ports and can use in multiple devices. Link aggregation allows increase bandwidth.

(wirelessbridge, 2023)

Router (Cisco ISR 4000 Series)

It uses Cisco Catalyst Software Defined WAN software features. It is high performance up to 2 Gbps throughput and used 4x GE RJ45 ports and - 4x GE SFP slots.

It is easier manageable and support. It is secure that it has Integrated threat defense, AES256 encryption and zone-based firewall.

It is easily increasing capacity by remote performance on demand license upgrade with no hardware upgrade. (Pay-as-you-grow)

(CISCO, 2024)

Switch (Cisco Catalyst 9300 Series)

It can up to 1 Tbps stackable switching bandwidth. It has a strong processor of 16GB Ram, external storage of 240 GB and can use IPv6.

It supports power and fan redundancy and uses StackPower. It protects device from hardware failure. It has strong security with features such as MACsec encryption, Secure Boot, TrustSec and Encrypted Traffic Analytics.

It can connect up to 8 switches in one stack and works efficiently in large network.

(CISCO, 2025)

Multilayer switch (Cisco MDS 9100 series)

It can support 64 Gbps across all 48 ports with non- blocking architecture and has low latency.

It is designed 99.999% uptime with hot-swappable power supplies and fans.

It has various physical ports (20, 40, 48) and provides scalability and flexibility.

(CISCO, 2025)

Access point (Ubiquiti UniFi U6)

It can up to 10.2 Gbps across 2.4 GHz, 5 GHz, and 6 GHz bands and uses new 6 GHz band. It is stable with high-speed connection.

It has strong wireless security (WPA3) and is designed for high-density usage. It provides continuous, uninterrupted wireless connection.

It can be installed simply and expanded well when more users are added. It is powered by 2.5GbE PoE+ and can reduce the number of cables needed and easier to expand the network.

(itman, 2025)

Ethernet cable (Cat6a ,10GBASE-T Ethernet) with RJ 45 Connector (T-568B)

It can up to 100 meters for 10 GbE and 500 MHz bandwidth. It supports high speed ethernet standards, clear signal transmission and communicate with no data loss.

It uses STP (Shielded Twisted Pair) shielding to reduce EMI (Electromagnetic Interference) and straight through cable. Signals are strong, stable and reliable.

It supports future technology and higher bandwidth needs. It is also compatible with many commonly used devices.

(ascentoptics, 2024)

Task 4

a) IP addressing plan

The network will use classless addressing (CIDR) and will be subnetted by VLSM (Variable Length Subnet Masking) for accurate and detailed subnetting.

For company which need to expand 300 employees in the future, Class B private IP range (eg.172.20.0.0/16) (255.255.0.0) will be used as it has about 65534 hosts and 16384 networks.

IP range (172.20.0.0/16) will be subnetted by VLSM into 28 networks and the remaining unused networks will be kept for the future growth.

Departments		Required Hosts	Needed VSLM	Usable total IPs
open plan workspace (1 st floor)	Development	20	/26	62
	Research	20	/26	62
	Customer service	20	/26	62
open plan workspace (3 rd floor)	Human resources	20	/26	62
	Finance	20	/26	62
	Sale and marketing	20	/26	62
dedicate spaces (IT support)		15	/27	30
common areas (break rooms, cafeteria)		5	/28	14
2 nd floor	4 conference rooms	20 (5 in each room)	/28	14
	Sever room	3	/29	6
	15 private offices (Administration)	15 (1 in each room)	/30	2

b) Subnet the network (VLSM)

Department	Subnet Assigned	Subnet Mask	Network IP	Usable IP	Broad cast IP
Development	172.20.0.0/26	255.255.255.192	172.20.0.0	172.20.0.1 to 172.20.0.62	172.20.0.63
Research	172.20.0.64/26	255.255.255.192	172.20.0.64	172.20.0.65 to 172.20.0.126	172.20.0.127
Customer service	172.20.0.128/26	255.255.255.192	172.20.0.128	172.20.0.129 to 172.20.0.190	172.20.0.191
Human resources	172.20.0.192/26	255.255.255.192	172.20.0.192	172.20.0.193 to 172.20.0.254	172.20.0.255
Finance	172.20.1.0/26	255.255.255.192	172.20.1.0	172.20.1.1 to 172.20.1.62	172.20.1.63
Sale and marketing	172.20.1.64/26	255.255.255.192	172.20.1.64	172.20.1.65 to 172.20.1.126	172.20.1.127
IT support	172.20.1.128/27	255.255.255.224	172.20.1.128	172.20.1.129 to 172.20.1.158	172.20.1.159
common areas	172.20.1.160	255.255.255.240	172.20.1.160	172.20.1.161 to 172.20.1.174	172.20.1.175
conference room 1	172.20.1.176	255.255.255.240	172.20.1.176	172.20.1.177 to 172.20.1.190	172.20.1.191
conference room 2	172.20.1.192	255.255.255.240	172.20.1.192	172.20.1.193 to 172.20.1.206	172.20.1.207
conference room 3	172.20.1.208	255.255.255.240	172.20.1.208	172.20.1.209 to 172.20.1.222	172.20.1.223
conference room 4	172.20.1.224	255.255.255.240	172.20.1.224	172.20.1.225 to 172.20.1.238	172.20.1.239
Sever room	172.20.1.240	255.255.255.248	172.20.1.240	172.20.1.241 to 172.20.1.246	172.20.1.247

Department	Subnet Assigned	Subnet Mask	Network IP	Usable IP	Broadcast IP
private office 1	172.20.1.248	255.255.255.252	172.20.1.248	172.20.1.249 to 172.20.1.250	172.20.1.251
private office 2	172.20.1.252	255.255.255.252	172.20.1.252	172.20.1.253 to 172.20.1.254	172.20.1.255
private office 3	172.20.2.0	255.255.255.252	172.20.2.0	172.20.2.1 to 172.20.2.2	172.20.2.3
private office 4	172.20.2.4	255.255.255.252	172.20.2.4	172.20.2.5 to 172.20.2.6	172.20.2.7
private office 5	172.20.2.8	255.255.255.252	172.20.2.8	172.20.2.9 to 172.20.2.10	172.20.2.11
private office 6	172.20.2.12	255.255.255.252	172.20.2.12	172.20.2.13 to 172.20.2.14	172.20.2.15
private office 7	172.20.2.16	255.255.255.252	172.20.2.16	172.20.2.17 to 172.20.2.18	172.20.2.19
private office 8	172.20.2.20	255.255.255.252	172.20.2.20	172.20.2.21 to 172.20.2.22	172.20.2.23
private office 9	172.20.2.24	255.255.255.252	172.20.2.24	172.20.2.25 to 172.20.2.26	172.20.2.27
private office 10	172.20.2.28	255.255.255.252	172.20.2.28	172.20.2.29 to 172.20.2.30	172.20.2.31
private office 11	172.20.2.32	255.255.255.252	172.20.2.32	172.20.2.33 to 172.20.2.34	172.20.2.35
private office 12	172.20.2.36	255.255.255.252	172.20.2.36	172.20.2.37 to 172.20.2.38	172.20.2.39
private office 13	172.20.2.40	255.255.255.252	172.20.2.40	172.20.2.41 to 172.20.2.42	172.20.2.43
private office 14	172.20.2.44	255.255.255.252	172.20.2.44	172.20.2.45 to 172.20.2.46	172.20.2.47
private office 15	172.20.2.48	255.255.255.252	172.20.2.48	172.20.2.49 to 172.20.2.50	172.20.2.51

c) future growth and scalability

In company, one private class B network (172.20.0.0/16) is subnetted into 28 networks. Each network uses only about 50% IPs for their hosts and has spare IPs for future growth. There are also about some IPs available in network that can be further subnetted for future devices and departments. Subnetting with VLSM can minimize waste and maximize space for expansion, future growth and scalability.

Task 5

a) Vulnerabilities and security threats

The common security threats are

1. Phishing and social engineering attacks

Attackers can easily attack human psychology than technological vulnerabilities. Dumpster diving, shoulder surfing techniques, phishing variants such as spear phishing, voice phishing, smishing are also used. Baiting such as USB baiting and pretexting are also used to infiltrate into the network system.

2. Malwares

Ransomwares, worms, virus, trojans and fileless malware infiltrates into network system and cause damage. Ransomwares with AI are also commonly used today.

3. DDoS attacks

Distributed denial of attack targets the availability of the network devices.

4. Man-in-the-middle attacks

Attackers stay in middle of communication and steal data or inject malicious data into network.

5. SQL injection

Attackers attack vulnerability of data base to extract information from network system.

6. IoT and OT threats

They are less secure and more vulnerable other devices in network. Supply chain attacks are also common due to insecure third-party vendors.

7. Zero-day exploits

Advanced targets are attacked by long term preparation. They use advance malwares, evasion methods and lateral movement in network system.

8. Insider threats

The risks came from people in company who make accidentally or intentionally data breached.

(Diego, 2025)

b) Security measures for threats

AI powered security operation center and automatic thread detection tools can be used to prevent AI powered attacks. For social engineering attack, employees in company should educated about security awareness and should use multifactor authentication.

IoT devices should be regular updating firmware, change default password, and network should be segmented. For supply chain attack, strict security standards must use in agreement contract.

For DDOS, utilizing rate limiting, anti DDOS software and hardware and web application firewall can be used. For Man-in-the-middle attack, protocols with SSL-TLS encryption should be used. For injection attack, validation should be added in input sources and data should be viewed and updated regularly.

To prevent malware threats, firewalls, antivirus, antimalware and advance threat detection system should be added. End users should also be educated and proper access control should be implemented. For insider threats, DLP and behavior analysis technologies should be added. The company should have clear policy for access control.

To prevent APTs, threat intelligence should be collected from various sources. Advanced detection technologies, encryption and segmentation with zero trust should be used. Regular assessment of security, continuous monitoring and updating should be done. Employees should be trained security awareness and should do incident response and forensic after attack was done.

(cynet, 2025)

Recommendation

Segmentation of network should be used to isolate lateral movement of attacks. VPN should be used especially for remote work. End point protection tools such as antivirus must be installed in system. Encryption should be necessary and system should update regularly to fix bugs.

Firewall with IDS must be used for prevention between trusted and untrusted network and protect unauthorized access for data breaches. Employee should be educated and access control management should be clearly determined. Multiple security tools should be added for maintaining visibility in complex network.

Task 6

a) Performance and network monitoring

Network monitoring tools (SolarWinds, PRTG, Zabbix, Nagios, OP Manger etc.), protocol analysis tools (Wireshark etc.), security monitoring tools (NGFW, Splunk, Cisco Stealth watch etc.) can be used to monitor network and security enhancement for company.

1. SolarWinds

It is used to monitor network traffic and bandwidth usage . Devices availability, servers and applications can be also monitored. (Nareshgologica, 2023)

2. PRTG

It is used to monitor network traffic and provides detail information of bandwidth usage, network performance and traffic analysis. (Squadcast, 2024)

3. Nagios

It can monitor memory usage, ram, log files, disk usage and microprocessor load and services such as SMTP, POP3, SSH, HTTP etc. (Ozturk, 2019)

4. Zabbix

Bandwidth usage, devices availability, performance and latency of network can be monitored. It can also monitor CPU load, memory usage and disk space. (Engineer, 2025)

5. Wireshark

It can detect data packets in real time. Protocols such as HTTP, UDP, FTP, DNS, TCP/IP etc. are supported for decoding. It can detect malicious activity and improving network efficiency. (Ayesha, 2024)

6. Next generation firewall

It is the combination of traditional firewall with DPI, IPS, SSL/TLS inspection, threat intelligence integration, user identity awareness, application awareness and control.

It can check contents of data and block malicious contents such as malware (even data is encrypted). It can determine role-based security policies. (Networking, 2024)

b) Network availability and reliability

For redundancy, backup switches, routers, servers should be added. Uninterruptible power supply (UPS) must be present for backup power supply in company. Backup internet connection and backup configuration settings should be present for restore availability in short time.

The system should have real time alerting system when one of devices is shut down and failover system to restore shut down device for high availability. For efficient internet connection, load balancing should be also needed to avoid overload.

Anti DDOS software and hardware should be used to prevent DDOS attacks. Network monitoring tools, protocol analysis tools and security monitoring tools should also be added into system for maintain availability and performance. Firmware and software also need to update regularly to get full security measures.

Conclusion

The company is provided detail LAN design with specific types of devices, number of employees and departments. It has also a plan for future growth with appropriate IPv4 addressing. Effective security measures and network monitoring tools are used to prevent the availability, integrity and confidentiality.

-----End of the Assignment-----

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